

BOL'SHOY SHANTAR, Russian Federation

WMO: 311740

Lat: **54.8623N**Lon: **137.4970E**Elev: **30**StdP: **14.68**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.5	-24.6	-32.5	0.9	-27.7	-28.7	1.1	-23.7	27.9	11.5	24.2	9.1	4.7	30	0.789

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.7	69.9	60.8	66.5	59.0	63.7	57.4	62.7	67.6	60.4	64.7	58.5	62.3	6.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.4	78.6	65.0	58.4	73.0	62.9	56.6	68.4	60.8	28.4	67.9	26.8	64.9	25.5	62.5	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	24.7	20.6	17.4	DB	-36.5	76.8	6.9	2.8	-41.5	78.9	-45.5	80.5	-49.4	82.1	-54.5	84.2
(5)				WB	-36.0	66.3	5.6	2.3	-40.1	68.0	-43.3	69.3	-46.5	70.6	-50.5	72.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	-4.3	-3.1	8.9	24.4	36.3	45.0	52.8	55.0	47.5	33.9	15.6	0.9
	DBStd	22.41	9.21	9.14	9.41	7.11	4.82	6.07	5.50	4.57	5.59	6.59	10.22	9.80
	HDD50	9021	1682	1488	1274	769	427	173	32	11	113	498	1031	1523
	HDD65	14152	2147	1908	1739	1219	891	599	379	312	526	963	1481	1988
	CDD50	345	0	0	0	0	1	24	119	164	37	0	0	0
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH74	22	0	0	0	0	0	0	14	8	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	7.2	7.3	7.2	7.4	7.0	6.9	5.4	5.0	5.5	7.0	8.9	10.0	8.9
Precipitation	PrecAvg	20.6	0.9	0.7	0.6	1.4	1.9	1.8	2.5	2.7	2.7	2.6	1.6	1.4
	PrecMax	30.3	2.7	2.2	3.9	3.7	4.7	5.6	6.5	9.6	6.2	9.1	6.4	4.9
	PrecMin	8.0	0.0	0.0	0.0	0.1	0.3	0.3	0.2	0.0	0.2	0.2	0.0	0.0
	PrecStd	5.3	0.8	0.6	0.8	0.9	1.2	1.1	1.8	2.0	1.7	1.9	1.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.6	25.5	36.5	45.1	60.1	70.0	75.3	74.1	67.6	55.1	39.5	30.5
		MCWB	26.1	23.8	30.8	37.2	47.7	58.2	64.3	63.8	58.3	46.2	34.8	29.3
	2%	DB	21.0	20.5	31.3	40.9	53.6	64.9	70.3	69.9	64.1	50.9	36.4	26.9
		MCWB	20.3	17.9	27.4	34.4	44.1	54.9	61.8	61.1	56.5	45.0	33.3	25.8
	5%	DB	14.6	16.1	28.0	37.5	49.0	60.7	65.8	66.7	61.5	47.9	33.4	19.5
		MCWB	13.1	14.1	25.1	32.5	41.6	52.8	59.7	59.6	54.5	42.7	31.1	18.2
	10%	DB	10.8	12.6	25.4	34.7	45.1	56.5	62.2	64.0	59.0	45.2	31.1	14.8
		MCWB	9.4	10.9	23.1	31.2	39.4	50.8	57.8	58.3	53.1	40.3	28.8	13.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.2	24.6	31.7	38.2	48.4	59.7	65.8	66.2	61.2	48.9	36.3	29.7
		MCDB	26.4	25.4	35.4	43.2	57.2	67.3	73.0	72.1	65.0	52.0	38.3	30.5
	2%	WB	20.4	18.2	27.9	35.1	44.8	56.2	63.0	63.0	58.0	46.2	33.7	25.6
		MCDB	21.2	20.0	30.4	39.5	52.7	63.2	68.4	67.5	62.3	49.7	35.7	26.8
	5%	WB	13.1	14.3	25.3	33.1	42.2	53.4	60.3	60.9	55.9	43.3	31.7	18.3
		MCDB	14.5	16.0	27.6	36.4	48.2	59.3	64.8	65.1	59.8	46.7	33.6	19.4
	10%	WB	9.5	11.0	23.1	31.7	39.8	50.9	58.0	59.2	54.0	41.4	28.9	13.4
		MCDB	10.7	12.6	25.5	34.3	44.4	56.0	61.7	62.7	57.8	44.4	30.9	14.6
Mean Daily Temperature Range	5% DB	MDBR	18.4	21.4	22.8	16.5	12.9	16.4	14.6	15.7	18.5	14.1	11.9	14.2
		MCDBR	16.2	20.5	22.8	19.9	23.2	26.2	22.4	21.8	22.3	19.8	11.9	14.1
		MCWBR	15.6	18.9	19.3	14.3	14.7	16.9	14.7	13.9	15.0	14.4	10.7	13.7
	5% WB	MCDBR	15.9	20.1	20.0	16.7	21.1	24.2	20.3	18.9	18.7	17.5	11.0	14.0
		MCWBR	15.4	18.4	17.2	12.5	13.9	16.8	14.1	13.4	13.9	13.7	10.1	13.7
Clear-Sky Solar Irradiance	taub	0.261	0.286	0.348	0.457	0.518	0.503	0.516	0.445	0.382	0.371	0.320	0.275	
	taud	2.044	2.018	1.868	1.648	1.671	1.934	2.016	2.243	2.360	2.174	2.106	2.018	
	Ebn at Noon	206	245	252	237	231	238	232	244	246	211	178	167	
	Edh at Noon	22	32	48	69	72	56	50	38	30	28	21	18	
All-Sky Solar Radiation	RadAvg	302	635	1088	1406	1401	1634	1453	1264	986	561	286	198	
	RadStd	23	55	114	140	267	180	196	164	92	61	35	27	

Historical Trends

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